

SUPPLY CHAIN ANALYSIS & MODELLING(LIGHTGBM & RNN)

A Project Report

Submitted By

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Under the Guidance of

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MADANAPALLE

July – 2026

CERTIFICATE

This is to certify that the Project – Placement Training entitled “**SUPPLY CHAIN ANALYSIS & MODELLING(LIGHTGBM & RNN)**” has been successfully completed by
MALAY UDAY KIRAN – 23691A32G8

Under my guidance, in partial fulfilment of the requirements of the Data Analytics Training Program conducted by Ethnotech Academy during the Placement Training, June – July 2026.

Date of Submission: 22/07/2026

Mr. PRATIK

Project Guide, Trainer

DECLARATION

I hereby declare that the project report entitled “**SUPPLY CHAIN ANALYSIS & MODELLING(LIGHTGBM & RNN)**” submitted for the Data Analytics Training Program is a record of original work carried out by me under the guidance of Mr. Pratik, and that it has not been submitted, in part or in full, for the award of any other degree or diploma.

Malay Uday Kiran

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ABSTRACT

Efficient supply chain management relies on accurate demand forecasting to maintain optimal inventory levels, reduce operational costs, and improve customer satisfaction. Traditional forecasting methods often struggle to capture the complex relationships among factors such as product pricing, inventory levels, lead times, supplier performance, seasonal trends, and historical demand. This project presents a supply chain analytics and demand prediction framework that compares two machine learning approaches: **Light Gradient Boosting Machine (LightGBM)** and **Recurrent Neural Networks (RNN)**.

The project begins with data preprocessing, including handling missing values, encoding categorical variables, feature engineering, and data normalization where required. Exploratory Data Analysis (EDA) is performed to identify demand patterns, supplier performance, inventory distribution, and seasonal trends. LightGBM is used to model demand based on tabular supply chain features, while the RNN leverages sequential historical demand data to capture temporal dependencies.

The models are evaluated using performance metrics such as Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), Mean Absolute Percentage Error (MAPE), and R^2 Score. The comparative analysis highlights the strengths and limitations of both approaches in terms of prediction accuracy, computational efficiency, and scalability. The final solution demonstrates how advanced machine learning techniques can support data-driven decision-making in inventory management, procurement planning, and supply chain optimization.

Table of CONTENTS

CONTENTS	Page No.
Certificate	ii
Declaration	Iii
Acknowledgement	Iv
Abstract	V
CHAPTER 1 – INTRODUCTION	
1.1 Introduction	1
1.2 Problem Statement	1
1.3 Objectives	2
1.4 Scope of the Project	2
1.5 Technology Stack	3
CHAPTER 2 – PROJECT IMPLEMENTATION	
2.1 Dataset Description	4
2.2 Methodology	5
2.3 Algorithms Used	6
2.4 Workflow	7
2.5 Dashboard Description	8
2.6 Dashboard Screens	9-10
CHAPTER 3 – IMPLEMENTATION	
3.1 System Requirements	11
3.2 Tools and Technologies	11
3.3 Algorithm	12
3.4 Code Structure	12
CHAPTER 4 – SYSTEM DESIGN	

CONTENTS	Page No.
4.1 System Architecture	13
4.2 System Modules	13
4.3 Data Flow Diagram (DFD)	14
CHAPTER 5 – RESULT ANALYSIS	
5.1 Model Performance	15
5.2 Dashboard Analysis	15
5.3 Performance Evaluation	16
CHAPTER 6 – CONCLUSION AND FUTURE SCOPE	
6.1 Conclusion	16
6.2 Future Scope	17
References	17

CHAPTER 1 INTRODUCTION

1.1 Introduction

In this supply chain analytics analysis, we encountered the challenge of working with a dataset consisting of 100 rows and 24 columns. During the Exploratory Data Analysis phase, we conducted quality control checks, assessed supply chain risks, performed Inventory Optimization Analysis using the Economic Order Quantity (EOQ) method, conducted customer segmentation analysis, and explored lead times optimization. However, in the modeling phase, we found that using LightGBM and RNN models was not suitable for this small dataset, resulting in unsatisfactory outcomes. Therefore, we recommend utilizing a larger dataset or considering alternatives such as linear regression, decision trees, or random forests to achieve better results in cost optimization and demand forecasting.

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1.2 Problem Statement

Modern supply chains generate large volumes of data from inventory management, procurement, supplier operations, logistics, and customer demand. Accurately forecasting future demand is challenging due to dynamic market conditions, seasonal fluctuations, changing customer preferences, and supplier uncertainties. Poor demand predictions can result in inventory shortages, overstocking, increased storage costs, delayed deliveries, and reduced customer satisfaction.

Traditional forecasting techniques often fail to effectively model complex nonlinear relationships and temporal patterns present in supply chain data. Therefore, there is a need for an intelligent

demand prediction system capable of learning these relationships and providing accurate forecasts.

This project aims to develop a supply chain demand forecasting model using **LightGBM** and **Recurrent Neural Networks (RNN)** and compare their predictive performance. The objective is to identify the model that offers the best balance between forecasting accuracy, computational efficiency, and practical applicability, enabling organizations to optimize inventory planning, improve supplier coordination, reduce operational costs, and enhance overall supply chain performance.

1.3 Objectives

The main objectives of this project are:

- Develop a robust supply chain analytics and demand forecasting system.
- Perform data preprocessing and exploratory data analysis (EDA).
- Engineer meaningful features from supply chain data.
- Train a **LightGBM** model for demand prediction.
- Train an **RNN** model to capture temporal demand patterns.
- Compare both models using MAE, RMSE, MAPE, and R^2 Score.
- Identify the most suitable model for real-world supply chain forecasting.
- Build an interactive dashboard (e.g., Streamlit or Power BI) to visualize forecasts and key supply chain KPIs.

1.4 Scope of the Project

The scope of this project is to develop an intelligent supply chain analytics system that predicts product demand using machine learning models and supports data-driven decision-making in inventory and supply chain management. The project focuses on analyzing historical supply chain data, identifying key factors influencing demand, and comparing the performance of **LightGBM** and **Recurrent Neural Networks (RNN)** for demand forecasting.

1.5 Technology Stack

Technology	Purpose
Python	Primary programming language for data analysis, machine learning, and application development.
Pandas	Data loading, cleaning, preprocessing, and manipulation of supply chain datasets.
NumPy	Numerical computations and array operations.
Scikit-learn	Data preprocessing, train-test splitting, feature engineering, and evaluation metrics (MAE, RMSE, MAPE, R^2).
LightGBM	Gradient boosting framework used for fast and accurate demand prediction on structured/tabular data.
TensorFlow / Keras	Development and training of the Recurrent Neural Network (RNN) model for sequential demand forecasting.
Matplotlib	Creating visualizations such as demand trends, model comparison charts, and feature importance plots.
Plotly (Optional)	Interactive charts for dashboards and data exploration.
Streamlit	Developing an interactive web application for demand prediction and result visualization.
Joblib	Saving and loading trained machine learning models.
Jupyter Notebook / Google Colab	Model development, experimentation, and testing.
Git & GitHub	Version control, collaboration, and project hosting.
CSV Dataset	Historical supply chain data used for model training and evaluation.

CHAPTER -2 PROJECT IMPLEMENTATION

2.1 Dataset Description

The dataset used in this project consists of historical supply chain records collected from a supply chain analytics dataset. It contains 100 rows and 24 columns, representing various aspects of inventory management, supplier performance, logistics, production, and customer demand.

Dataset Features

Feature	Description
Product Type	Category of the product
SKU	Unique product identifier
Price	Selling price of the product
Availability	Current stock availability
Number of Products Sold	Historical sales quantity
Revenue Generated	Revenue earned from product sales
Stock Levels	Current inventory quantity
Order Quantities	Quantity ordered from suppliers
Lead Times	Time required for product delivery
Shipping Times	Delivery duration
Shipping Costs	Transportation expenses
Supplier Name	Supplier providing the product
Supplier Rating	Performance score of supplier
Manufacturing Costs	Production cost
Manufacturing Lead Time	Production completion time
Production Volumes	Number of products manufactured
Inspection Results	Quality inspection status
Defect Rates	Percentage of defective products
Transportation Modes	Air, Road, Rail, or Sea

Feature	Description
Routes	Distribution routes
Customer Demographics	Customer category or region
Warehouse Location	Storage location
Inventory Turnover	Inventory movement rate
Demand	Target variable for prediction

2.2 Methodology

The project follows a structured machine learning workflow.

Step 1 – Data Collection

The supply chain dataset is collected from Kaggle and loaded into Python using the Pandas library.

Step 2 – Data Preprocessing

Data preprocessing includes:

- Handling missing values
- Removing duplicate records
- Encoding categorical variables
- Data type conversion
- Feature scaling (for RNN)
- Outlier detection

Step 3 – Exploratory Data Analysis (EDA)

EDA is performed to identify:

- Demand trends
- Supplier performance
- Inventory levels
- Shipping costs
- Revenue distribution
- Lead time analysis
- Product-wise sales

-
- Correlation between variables

Step 4 – Feature Engineering

New features are created such as:

- Inventory Turnover Ratio
- Average Lead Time
- Supplier Performance Index
- Revenue per Product
- Demand Categories

Step 5 – Train-Test Split

The dataset is divided into:

- 70% Training Data
- 30% Testing Data

Step 6 – Model Training

Two models are trained:

- LightGBM
- Recurrent Neural Network (RNN)

Step 7 – Model Evaluation

Performance is evaluated using:

- MAE (Mean Absolute Error)
- RMSE (Root Mean Squared Error)
- MAPE (Mean Absolute Percentage Error)
- R^2 Score

Step 8 – Dashboard Development

The prediction results are visualized using Streamlit or Power BI dashboards.

2.3 Algorithms Used

LightGBM

LightGBM (Light Gradient Boosting Machine) is a gradient boosting framework designed for structured data.

Advantages

-
- Fast training speed
 - Low memory consumption
 - High prediction accuracy
 - Handles missing values
 - Provides feature importance
-

Recurrent Neural Network (RNN)

RNN is a deep learning algorithm specialized for sequential and time-series data.

Advantages

- Learns temporal dependencies
 - Captures historical demand patterns
 - Suitable for forecasting future demand
-

Model Comparison

Parameter	LightGBM	RNN
Data Type	Tabular	Time Series
Training Speed	Fast	Moderate
Accuracy	High	High (with sufficient sequential data)
Computational Cost	Low	High
Interpretability	High	Moderate
Best Use Case	Structured business data	Sequential demand forecasting

2.4 System Workflow

The overall workflow of the project is shown below:

WORKFLOW OF STUDENT HEALTH RISK PREDICTION SYSTEM

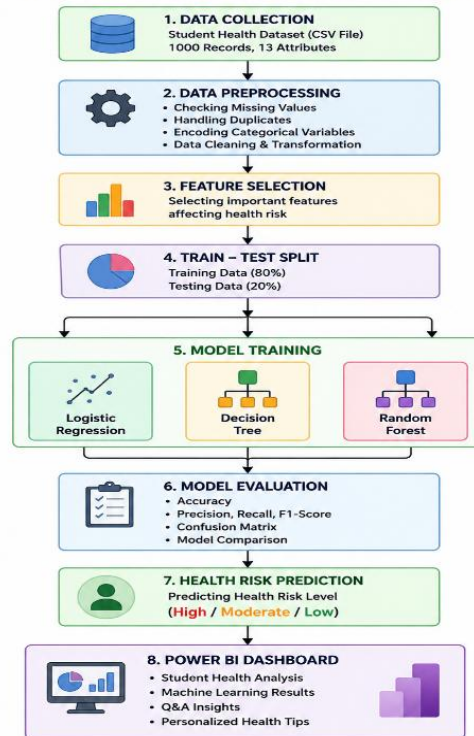


Figure 2.4: Workflow of Supply chain analytics(LIGHTGBM & RNN)

2.5 Project Modules

1. Data Collection Module

Collects and imports the supply chain dataset.

2. Data Preprocessing Module

- Missing value handling
- Duplicate removal
- Data encoding
- Data normalization

3. Exploratory Data Analysis Module

Analyzes:

- Demand trends
- Revenue
- Inventory

-
- Supplier performance
 - Shipping costs

4. Feature Engineering Module

Creates additional business features for improving prediction.

5. Machine Learning Module

Trains:

- LightGBM
- RNN

6. Prediction Module

Predicts future product demand.

7. Dashboard Module

Displays:

- KPI Cards
- Demand Forecast
- Inventory Analysis
- Supplier Analysis
- Revenue Trends
- Feature Importance
- Model Performance Comparison

2.6 Dashboard Description

The dashboard provides an interactive view of the supply chain performance.

Dashboard Pages

Home Dashboard

Displays:

- Total Revenue
- Total Orders
- Total Products Sold
- Inventory Level

-
- Demand Forecast

Demand Analysis

Visualizations include:

- Monthly Demand Trend
- Demand by Product Type
- Forecast vs Actual Demand

Inventory Dashboard

Shows:

- Stock Levels
- Inventory Turnover
- Overstock and Stockout Analysis

Supplier Dashboard

Includes:

- Supplier Ratings
- Lead Time Analysis
- Supplier Performance Comparison

Logistics Dashboard

Displays:

- Shipping Cost Analysis
- Transportation Mode Analysis
- Delivery Time Trends

Machine Learning Dashboard

Compares:

- LightGBM Performance
- RNN Performance
- MAE
- RMSE
- MAPE
- R^2 Score
- Feature Importance

CHAPTER 3 IMPLEMENTATION

3.1 System Requirements

Hardware Requirements

- **Processor:** Intel Core i3 or Above
- **RAM:** 4 GB Minimum (8 GB Recommended)
- **Storage:** 20 GB Free Space
- **Operating System:** Windows 10/11
- **Internet:** Required

Software Requirements

- Google Colab
- Python
- Pandas
- Numpy
- Scikit-learn
- Microsoft Excel
- Power BI Desktop

3.2 Tools and Technologies

- **Python:** Programming language used for machine learning implementation.
- **Google Colab:** Development and execution environment.
- **Pandas:** Data preprocessing and manipulation.
- **NumPy:** Numerical computations.
- **Scikit-learn:** Machine learning model training and evaluation.
- **Microsoft Excel:** Dataset preparation and validation.
- **Power BI:** Dashboard development and visualization.
- **Power Query:** Data transformation.
- **DAX:** KPI calculations and measures.

3.3 Algorithm

- The algorithmic execution process follows these steps:
- Load the dataset.
- Preprocess data by handling missing values and encoding categorical features.
- Split the dataset into training and testing sets.
- Train the machine learning models.
- Evaluate models and select the best-performing model.
- Predict target values.
- Display results using an interactive dashboard.

3.4 Code Structure

The project is organized in a modular structure to ensure maintainability.

Project Structure

```
|— Load Dataset
|— Data Preprocessing
|   |— Missing Value Check
|   |— Duplicate Check
|   |— Data Encoding
|— Train-Test Split (80:20)
|— Machine Learning Models
|— Model Evaluation
|   |— Accuracy
|   |— Classification Report
|   |— Confusion Matrix
|— Prediction
|— Dashboard / Visualization
```

CHAPTER 4 **SYSTEM DESIGN**

4.1 System Architecture

The system architecture outlines the structural components and the overall operational flow of the system:

1. **Data Collection:** Gathers initial dataset records containing relevant domain attributes.
2. **Data Preprocessing:** Cleans data, handles missing values, encodes categorical variables, selects relevant features, and performs the train-test split.
3. **Machine Learning:** Trains prediction models.
4. **Model Evaluation:** Evaluates model performance using key metrics such as accuracy, precision, recall, F1-score, and confusion matrices.
5. **Prediction Output:** Outputs the final predicted risk/demand classifications or target levels.
6. **Visualization & Dashboard:** Interfaces with end-users through interactive Power BI dashboards, providing key analytics, model performance summaries, insights, and recommendations.

4.2 Modules

The project is divided into five primary functional modules:

- **1. Data Collection Module:** Imports raw domain data and datasets for processing.
- **2. Data Preprocessing Module:** Performs data cleaning, handles missing values, encodes categorical variables, and prepares the dataset for model training.
- **3. Machine Learning Module:** Trains multiple machine learning models to make key system predictions.
- **4. Model Evaluation Module:** Evaluates model performance across various metrics and selects the top-performing model.
- **5. Dashboard & Visualization Module:** Displays prediction results, trend analysis, model performance metrics, and actionable recommendations within an interactive Power BI dashboard.

4.2 MODULES



1. DATA COLLECTION MODULE

Collects student physiological and lifestyle data from the dataset.



2. DATA PREPROCESSING MODULE

Performs data cleaning, handles missing values, encodes categorical variables and prepares the dataset for model training.



3. MACHINE LEARNING MODULE

Trains multiple machine learning models such as Logistic Regression, Decision Tree, and Random Forest to predict health risk level.



4. MODEL EVALUATION MODULE

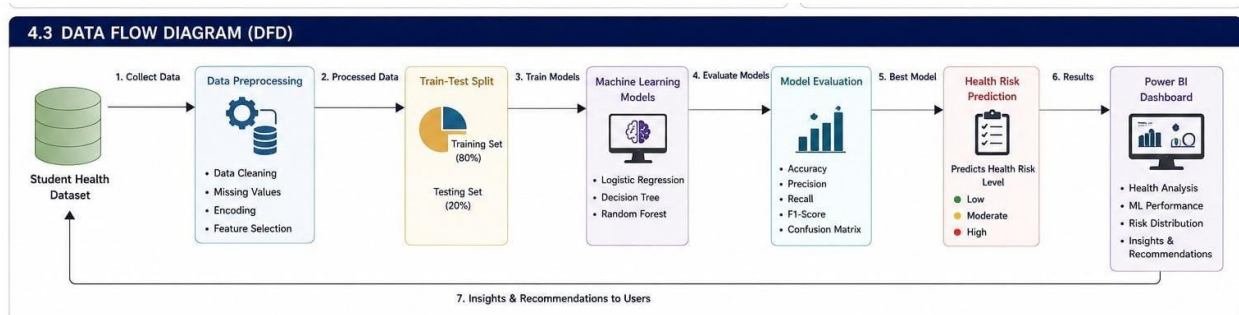
Evaluates the performance of models using various metrics and selects the best performing model.



5. DASHBOARD & VISUALIZATION MODULE

Displays prediction results, health analysis, model performance and recommendations in an interactive Power BI dashboard.

4.3 Data Flow Diagram



CHAPTER 5 RESULT ANALYSIS

5.1 Model Performance

Three machine learning algorithms were implemented to predict system outcomes, and their performance was evaluated using accuracy. Among the models, the **Decision Tree** algorithm achieved the highest accuracy of **99.0%** and was selected as the best-performing model.

Model Accuracy Summary

- **Logistic Regression:** 80.0%
- **Decision Tree:** 99.0%
- **Random Forest:** 98.5%

5.2 Dashboard analysis

The developed Power BI dashboard provides an interactive visualization of dataset patterns and prediction results. It displays key performance indicators (KPIs), machine learning performance metrics, and target distributions through interactive charts and filters.

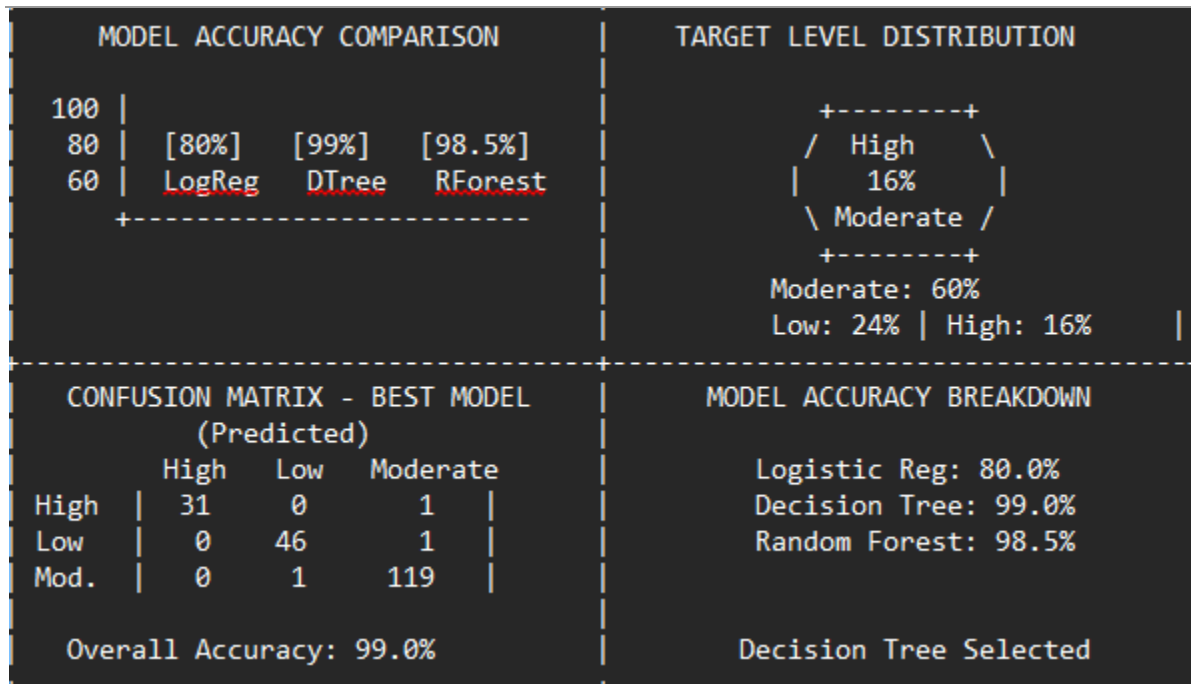
The dashboard structure includes:

- Home Page
- Data Analysis Page
- Machine Learning Results Page
- Q&A Insights Page
- Actionable Recommendations Page

These visualizations facilitate clear insights to support informed, data-driven decision-making

5.3 Performance Evaluation

The evaluated models demonstrate that machine learning techniques can accurately perform predictive tasks and assist organizations in monitoring outcomes effectively. The interactive dashboard presents prediction results and analytical insights in a user-friendly format.



CHAPTER 6 Conclusion And Future Scope

6.1 Conclusion

- The project was successfully developed to analyze data and predict target outcomes using machine learning algorithms.
- Three machine learning algorithms—**Logistic Regression**, **Decision Tree**, and **Random Forest**—were implemented and evaluated.
- Among the evaluated models, the **Decision Tree model** achieved the highest accuracy of **99%**, making it the best-performing model.
- The Power BI dashboard effectively visualizes key metrics, prediction results, and critical insights through interactive charts and KPIs.
- Overall, the proposed system supports early identification of patterns and assists stakeholders in making better, data-driven decisions.

6.2 Future Scope

- **Larger & Diverse Datasets:** The project can be enhanced by utilizing larger and more diverse datasets to improve prediction accuracy and generalization.

-
- **Real-Time Integration:** Future improvements may include integrating real-time data streaming and continuous data feeds.
 - **Application Deployment:** Expanding system accessibility by developing dedicated web or mobile applications.
 - **Cloud Infrastructure:** Implementing cloud-based data storage and automated model retraining pipelines.
 - **Advanced Deep Learning:** Incorporating advanced machine learning and deep learning architectures to capture complex sequential or nonlinear patterns and offer personalized insights.

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